In [ ]:

Task 5: Exploratory Data Analysis

Objective: Identify patterns, relationships, trends, and anomalies using statistical analysis and visualizations.

Tools: Python, Pandas, Matplotlib, Seaborn

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
```

In [2]:

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Sample - Superstore.csv to Sample - Superstore.csv
```

In [3]:

```
df = pd.read_csv('Sample - Superstore.csv', encoding='latin1')

df.head()
```

Out[3]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale

5 rows × 21 columns

In [4]:

```
print("Dataset shape:", df.shape)
df.info()
```

```
Dataset shape: (9994, 21)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Row ID         9994 non-null   int64  
 1   Order ID       9994 non-null   object 
 2   Order Date     9994 non-null   object 
 3   Ship Date      9994 non-null   object 
 4   Ship Mode      9994 non-null   object 
 5   Customer ID    9994 non-null   object 
 6   Customer Name  9994 non-null   object 
 7   Segment        9994 non-null   object 
 8   Country        9994 non-null   object 
 9   City           9994 non-null   object 
 10  State          9994 non-null   object 
 11  Postal Code    9994 non-null   int64  
 12  Region         9994 non-null   object 
 13  Product ID     9994 non-null   object 
 14  Category       9994 non-null   object 
 15  Sub-Category   9994 non-null   object 
 16  Product Name   9994 non-null   object 
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64  
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

In [5]:

```
df.describe(include="all").T
```

Out[5]:

	count	unique	top	freq	mean	std	min	25%	50%	
Row ID	9994.0	NaN	NaN	NaN	4997.5	2885.163629	1.0	2499.25	4997.5	7
Order ID	9994	5009	CA-2017-100111	14	NaN	NaN	NaN	NaN	NaN	
Order Date	9994	1237	9/5/2016	38	NaN	NaN	NaN	NaN	NaN	

	count	unique	top	freq	mean	std	min	25%	50%	
Ship Date	9994	1334	12/16/2015	35	NaN	NaN	NaN	NaN	NaN	
Ship Mode	9994	4	Standard Class	5968	NaN	NaN	NaN	NaN	NaN	
Customer ID	9994	793	WB-21850	37	NaN	NaN	NaN	NaN	NaN	
Customer Name	9994	793	William Brown	37	NaN	NaN	NaN	NaN	NaN	
Segment	9994	3	Consumer	5191	NaN	NaN	NaN	NaN	NaN	
Country	9994	1	United States	9994	NaN	NaN	NaN	NaN	NaN	
City	9994	531	New York City	915	NaN	NaN	NaN	NaN	NaN	
State	9994	49	California	2001	NaN	NaN	NaN	NaN	NaN	
Postal Code	9994.0	NaN	NaN	NaN	55190.379428	32063.69335	1040.0	23223.0	56430.5	9
Region	9994	4	West	3203	NaN	NaN	NaN	NaN	NaN	
Product ID	9994	1862	OFF-PA-10001970	19	NaN	NaN	NaN	NaN	NaN	
Category	9994	3	Office Supplies	6026	NaN	NaN	NaN	NaN	NaN	
Sub-Category	9994	17	Binders	1523	NaN	NaN	NaN	NaN	NaN	
Product Name	9994	1850	Staple envelope	48	NaN	NaN	NaN	NaN	NaN	
Sales	9994.0	NaN	NaN	NaN	229.858001	623.245101	0.444	17.28	54.49	
Quantity	9994.0	NaN	NaN	NaN	3.789574	2.22511	1.0	2.0	3.0	
Discount	9994.0	NaN	NaN	NaN	0.156203	0.206452	0.0	0.0	0.2	
Profit	9994.0	NaN	NaN	NaN	28.656896	234.260108	-6599.978	1.72875	8.6665	

In [6]:

```
df.nunique().sort_values()
```

Out[6]:

	0
Country	1
Segment	3
Category	3
Ship Mode	4
Region	4
Discount	12
Quantity	14

	0
Sub-Category	17
State	49
City	531
Postal Code	631
Customer Name	793
Customer ID	793
Order Date	1237
Ship Date	1334
Product Name	1850
Product ID	1862
Order ID	5009
Sales	5825
Profit	7287
Row ID	9994

dtype: int64

In [7]:

```
print("Duplicate rows:", df.duplicated().sum())

missing = df.isnull().sum()
missing_percentage = (missing / len(df)) * 100

pd.DataFrame({
    "Missing Values": missing,
    "Percentage": missing_percentage
}).sort_values("Missing Values", ascending=False)
```

```
Duplicate rows: 0
```

Out[7]:

	Missing Values	Percentage
Row ID	0	0.0
Order ID	0	0.0
Order Date	0	0.0
Ship Date	0	0.0
Ship Mode	0	0.0
Customer ID	0	0.0
Customer Name	0	0.0
Segment	0	0.0
Country	0	0.0
City	0	0.0

	Missing Values	Percentage
State	0	0.0
Postal Code	0	0.0
Region	0	0.0
Product ID	0	0.0
Category	0	0.0
Sub-Category	0	0.0
Product Name	0	0.0
Sales	0	0.0
Quantity	0	0.0
Discount	0	0.0
Profit	0	0.0

In [8]:

```
df = df.drop_duplicates()
```

Observation

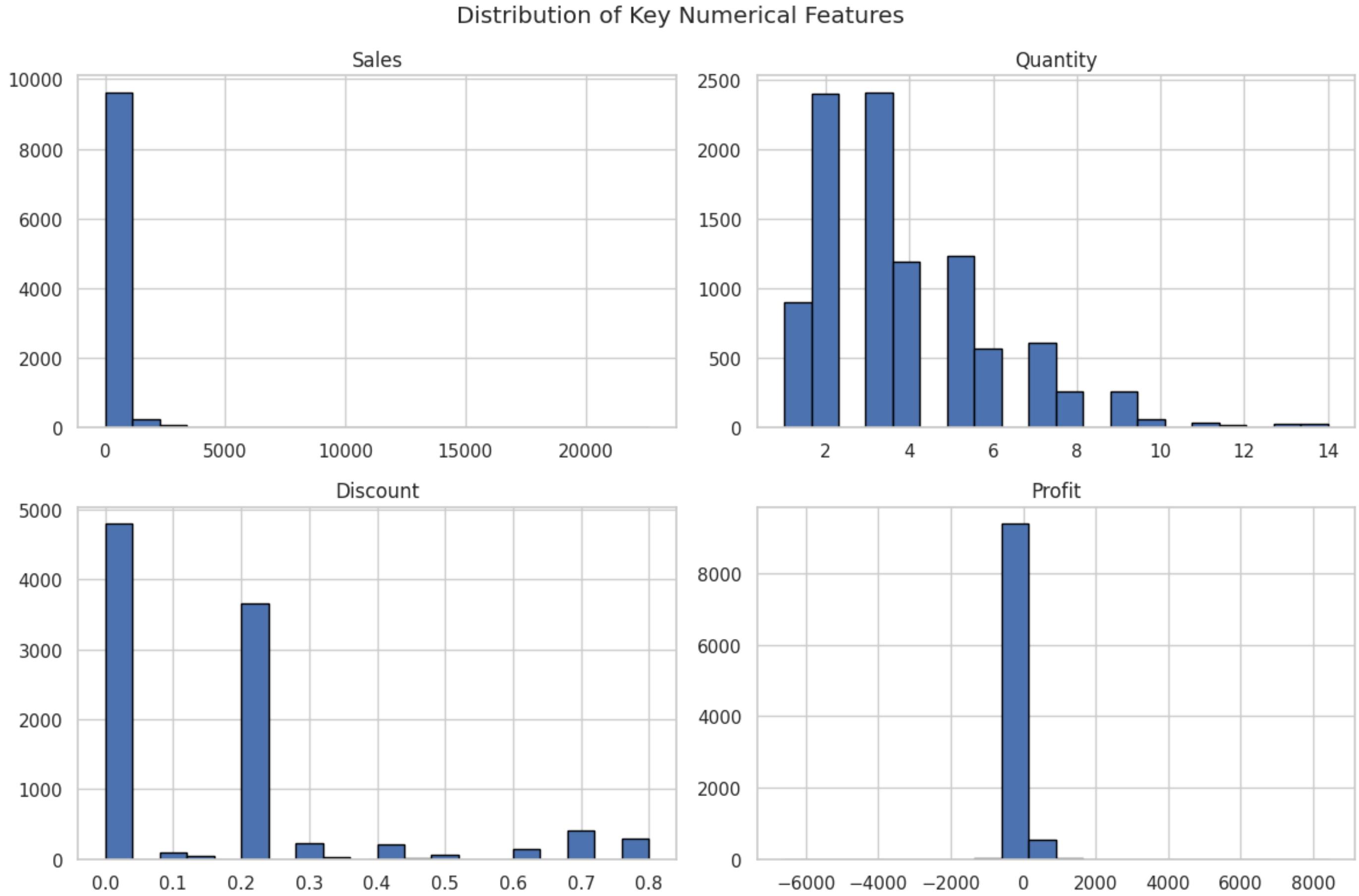
- The dataset contains 9,994 rows and 21 columns.
- It includes both numerical and categorical features.
- Sales, Profit, Quantity, and Discount are the main numerical variables.

In [22]:

```
numeric_columns = ['Sales', 'Quantity', 'Discount', 'Profit']

df[numeric_columns].hist(
    figsize=(12,8),
    bins=20,
    edgecolor='black'
)

plt.suptitle('Distribution of Key Numerical Features')
plt.tight_layout()
plt.show()
```

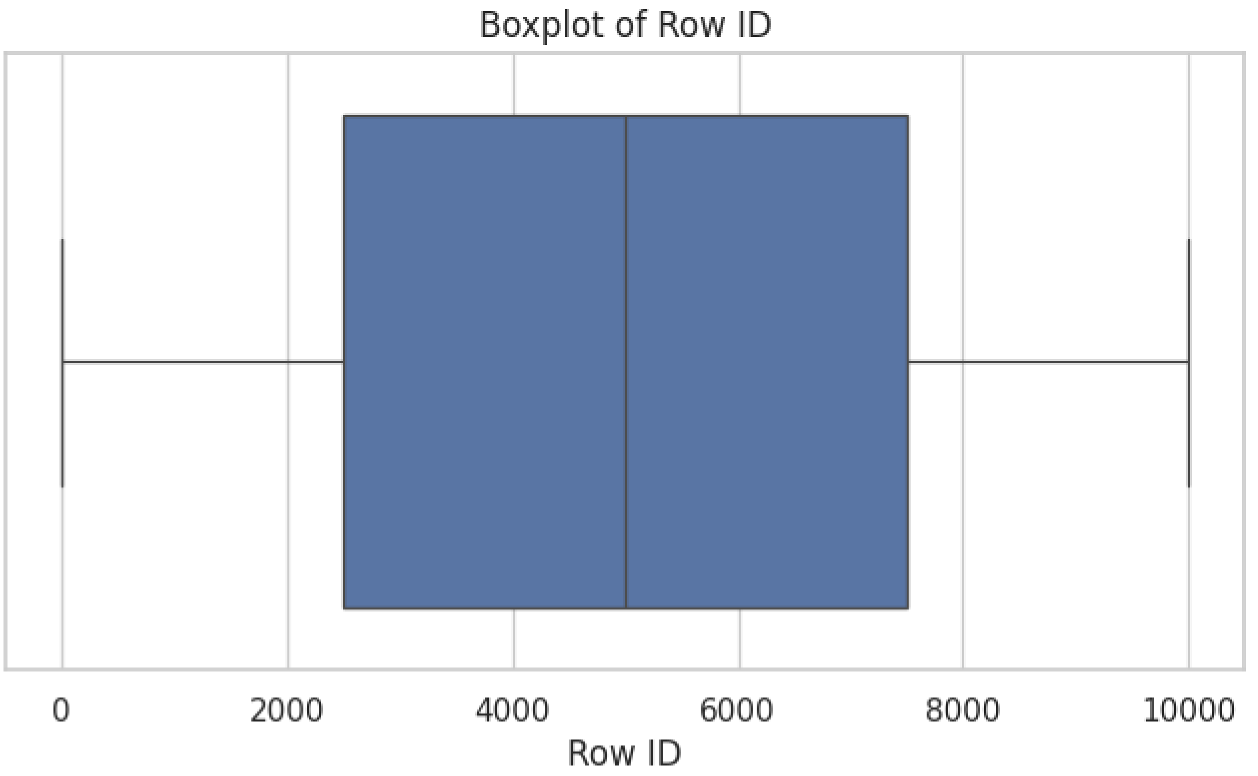


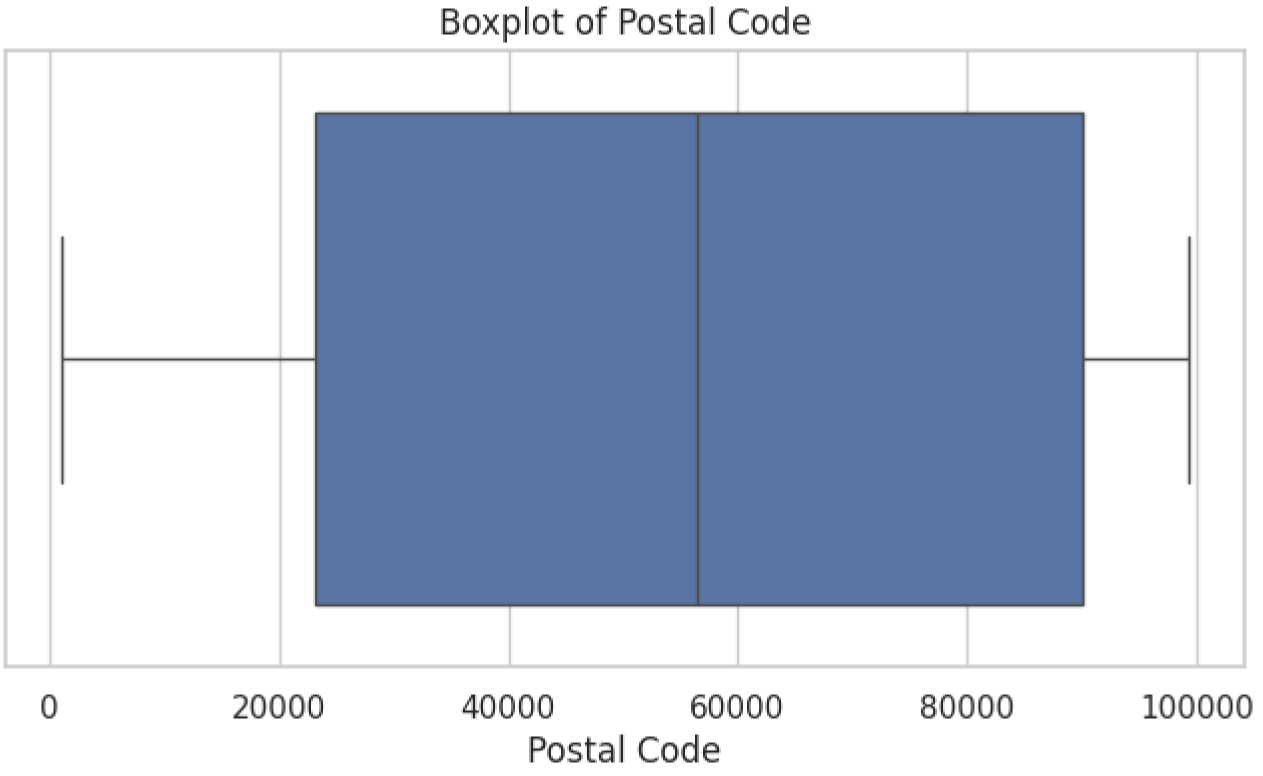
Observation

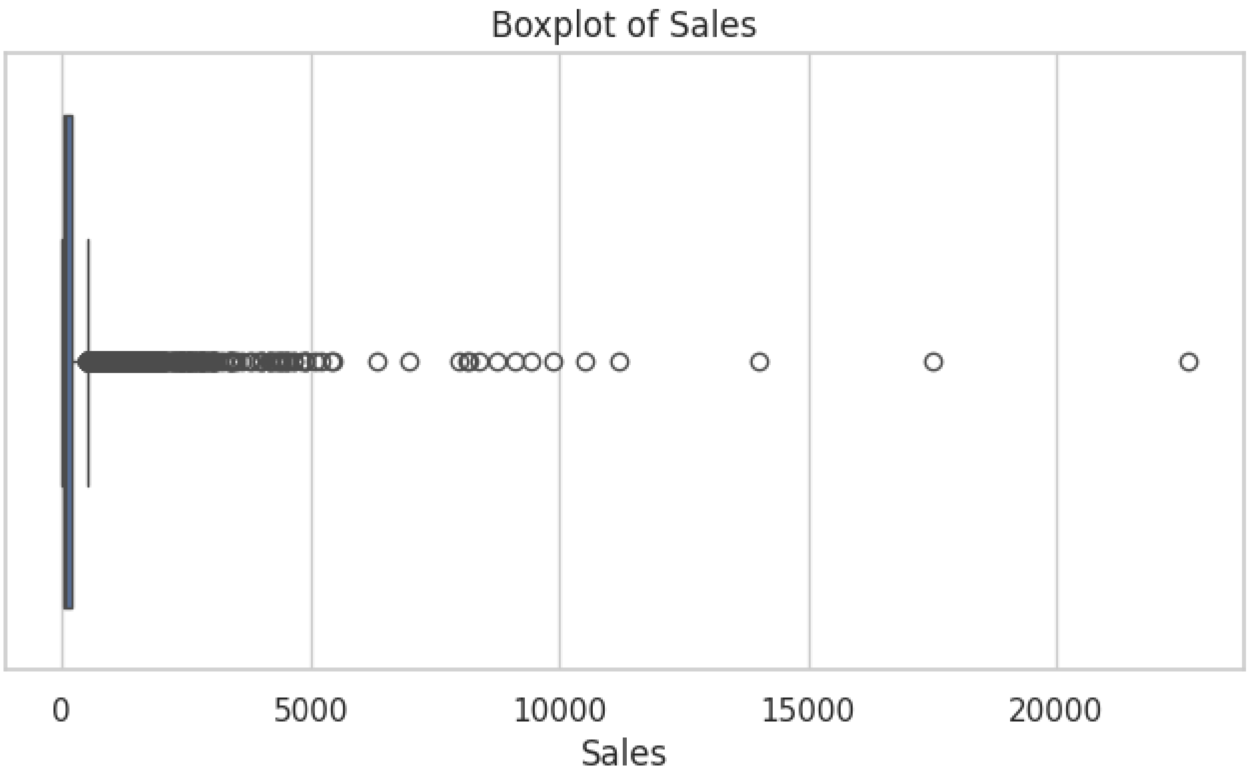
- Sales distribution is right-skewed, indicating that most orders have low sales while a few orders have very high sales.
- Profit shows significant variation, with some transactions resulting in losses.
- Most orders contain small quantities of products.
- Discounts are generally low, with higher discounts occurring less frequently.

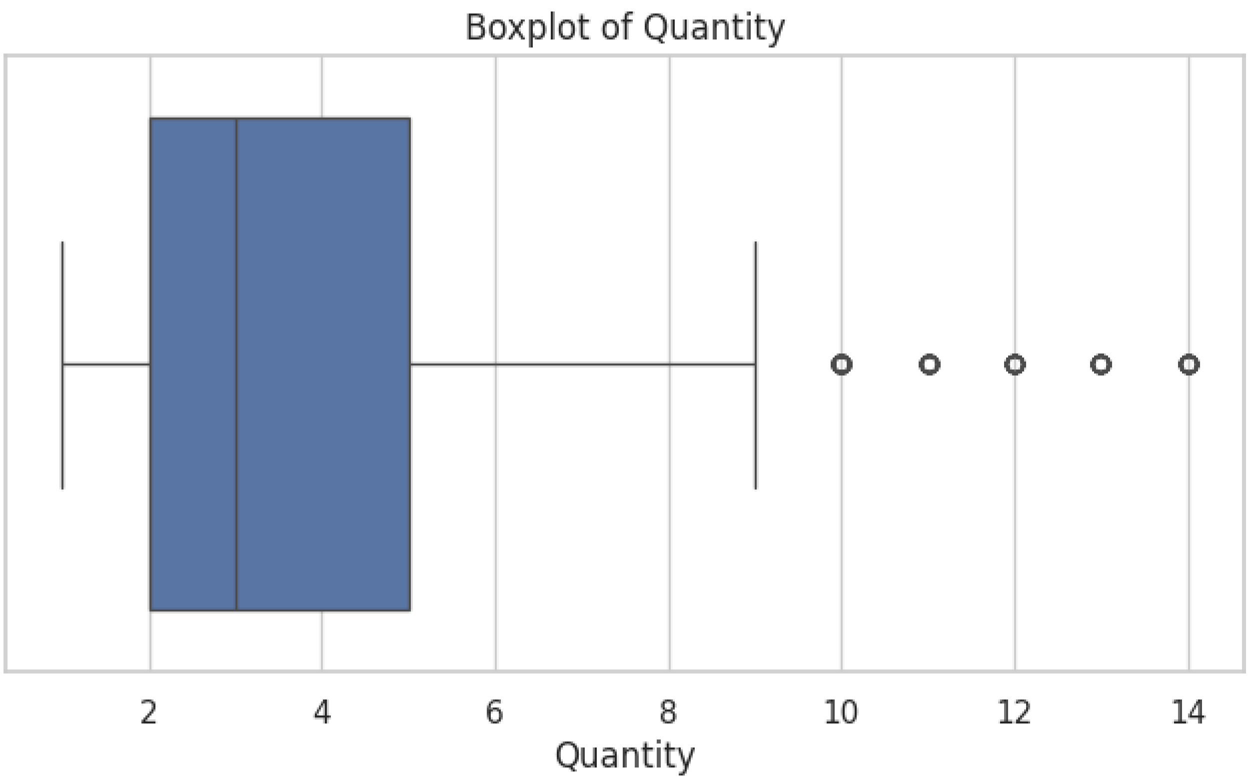
In [10]:

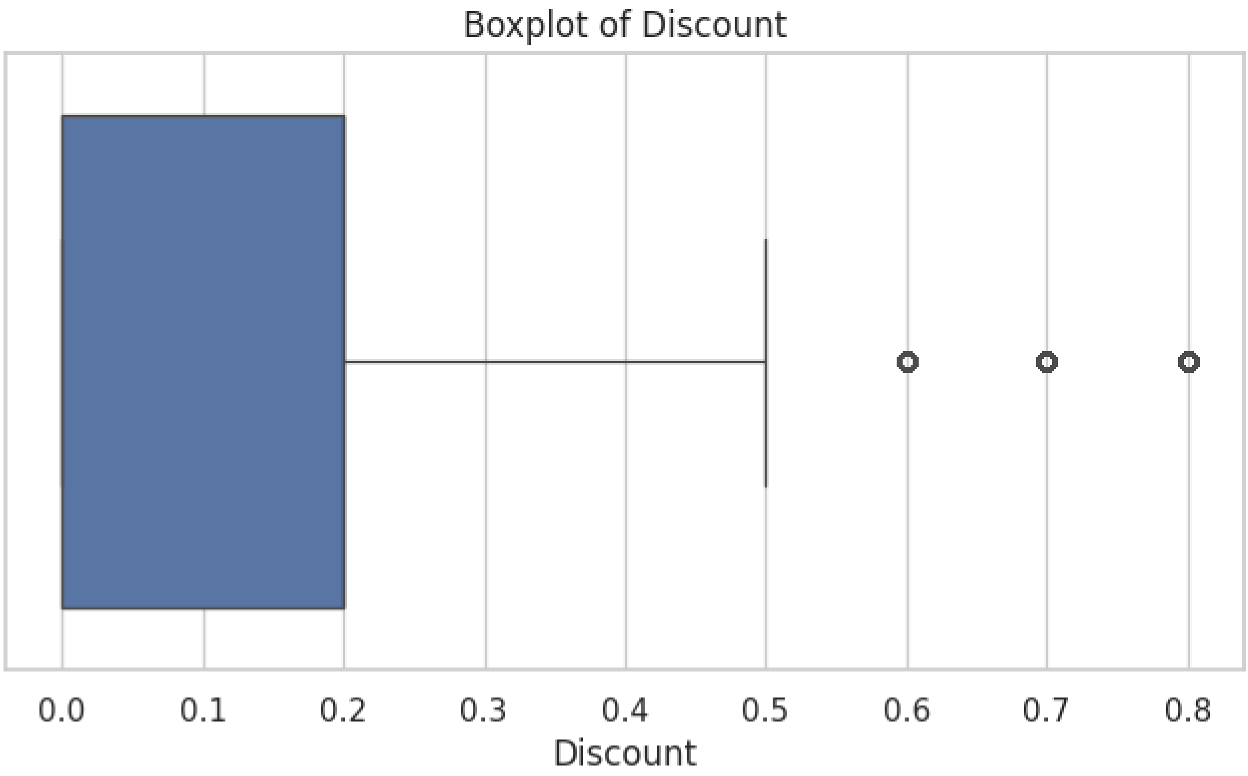
```
for column in numeric_columns:
    plt.figure(figsize=(8, 4))
    sns.boxplot(x=df[column])
    plt.title(f"Boxplot of {column}")
    plt.show()
```

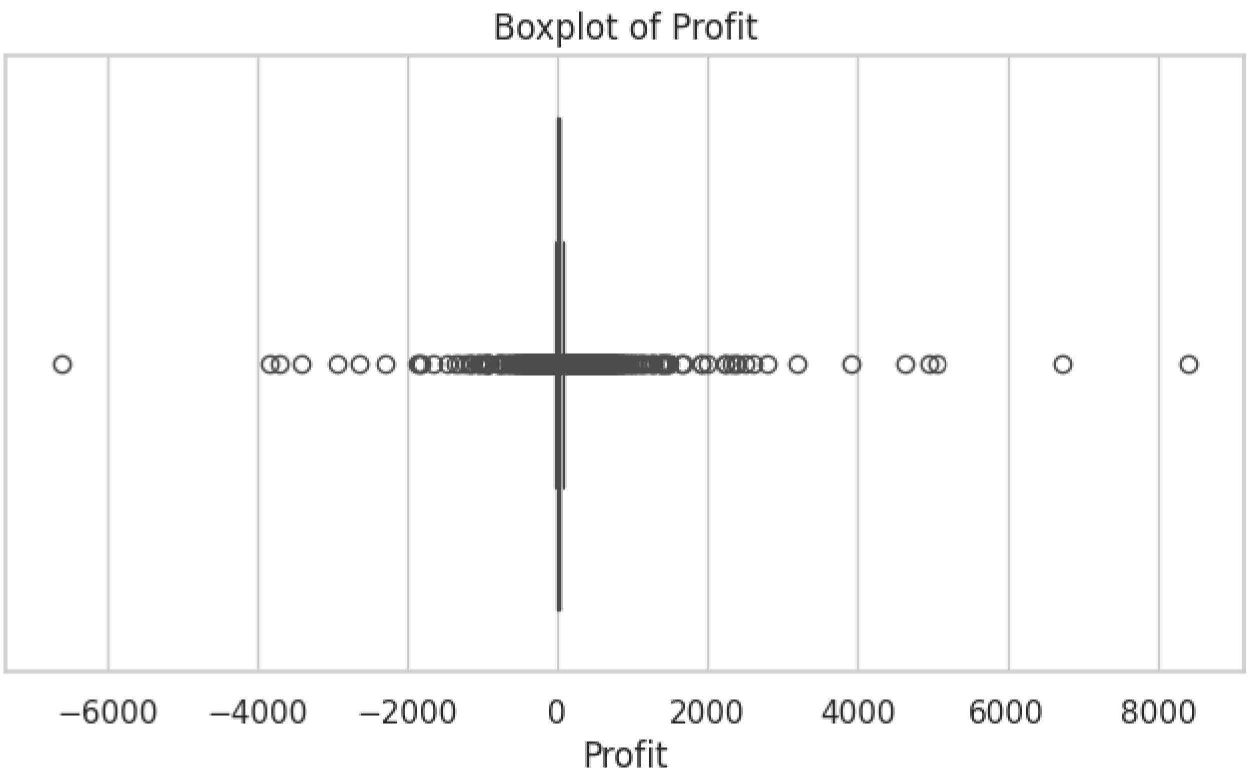

Boxplot of Row ID
0
2000
4000
6000
8000
10000
Row ID


Boxplot of Postal Code
0
20000
40000
60000
80000
100000
Postal Code


Boxplot of Sales
0
5000
10000
15000
20000
Sales


Boxplot of Quantity
2
4
6
8
10
12
14
Quantity





Observation

- Sales contains a large number of outliers, indicating a few exceptionally high-value transactions compared to the majority of orders.
- Quantity shows a small number of outliers, suggesting that only a few orders involve unusually large quantities.
- Discount contains several high-value outliers (60%, 70%, and 80%), indicating that large discounts are offered infrequently.

- Most orders have relatively low sales values, moderate quantities, and discounts below 20%.
- Row ID and Postal Code are identifier fields and do not provide meaningful business insights for outlier analysis.
- The presence of outliers in Sales, Quantity, and Discount should be considered during further analysis and predictive modeling.

In [11]:
```
categorical_columns = df.select_dtypes(include=["object", "category"]).columns

for column in categorical_columns:
    print(f"\nValue counts for {column}:")
    display(df[column].value_counts().head(10))
```
```
Value counts for Order ID:
```

Order ID	count
CA-2017-100111	14
CA-2017-157987	12
CA-2016-165330	11
US-2016-108504	11
US-2015-126977	10
CA-2016-105732	10
CA-2015-131338	10
CA-2015-158421	9
CA-2014-106439	9
US-2015-163433	9

dtype: int64

```
Value counts for Order Date:
```

Order Date	count
9/5/2016	38
9/2/2017	36
11/10/2016	35
12/1/2017	34
12/2/2017	34
12/9/2017	33
11/12/2017	30
12/8/2017	30
9/9/2017	29

Order Date	count
9/4/2017	28

dtype: int64

```
Value counts for Ship Date:
```

Ship Date	count
12/16/2015	35
9/26/2017	34
11/21/2017	32
12/6/2017	32
9/6/2017	30
12/12/2017	30
9/15/2017	30
9/13/2014	27
9/8/2017	27
9/26/2015	26

dtype: int64

```
Value counts for Ship Mode:
```

Ship Mode	count
Standard Class	5968
Second Class	1945
First Class	1538
Same Day	543

dtype: int64

```
Value counts for Customer ID:
```

Customer ID	count
WB-21850	37
MA-17560	34
JL-15835	34
PP-18955	34
CK-12205	32

Customer ID	count
JD-15895	32
EH-13765	32
SV-20365	32
ZC-21910	31
EP-13915	31

dtype: int64

```
Value counts for Customer Name:
```

Customer Name	count
William Brown	37
Matt Abelman	34
John Lee	34
Paul Prost	34
Chloris Kastensmidt	32
Jonathan Doherty	32
Edward Hooks	32
Seth Vernon	32
Zuschuss Carroll	31
Emily Phan	31

dtype: int64

```
Value counts for Segment:
```

Segment	count
Consumer	5191
Corporate	3020
Home Office	1783

dtype: int64

```
Value counts for Country:
```

Country	count
United States	9994

dtype: int64

```
Value counts for City:
```

City	count
New York City	915
Los Angeles	747
Philadelphia	537
San Francisco	510
Seattle	428
Houston	377
Chicago	314
Columbus	222
San Diego	170
Springfield	163

dtype: int64

```
Value counts for State:
```

State	count
California	2001
New York	1128
Texas	985
Pennsylvania	587
Washington	506
Illinois	492
Ohio	469
Florida	383
Michigan	255
North Carolina	249

dtype: int64

```
Value counts for Region:
```

Region	count
West	3203
East	2848

Region	count
Central	2323
South	1620

dtype: int64

```
Value counts for Product ID:
```

Product ID	count
OFF-PA-10001970	19
TEC-AC-10003832	18
FUR-FU-10004270	16
FUR-CH-10002647	15
FUR-CH-10001146	15
TEC-AC-10002049	15
TEC-AC-10003628	15
OFF-BI-10002026	14
FUR-CH-10002880	14
FUR-FU-10001473	14

dtype: int64

```
Value counts for Category:
```

Category	count
Office Supplies	6026
Furniture	2121
Technology	1847

dtype: int64

```
Value counts for Sub-Category:
```

Sub-Category	count
Binders	1523
Paper	1370
Furnishings	957
Phones	889
Storage	846

Sub-Category	count
Art	796
Accessories	775
Chairs	617
Appliances	466
Labels	364

dtype: int64

```
Value counts for Product Name:
```

Product Name	count
Staple envelope	48
Staples	46
Easy-staple paper	46
Avery Non-Stick Binders	20
Staples in misc. colors	19
KI Adjustable-Height Table	18
Staple remover	18
Storex Dura Pro Binders	17
Staple-based wall hangings	16
Situations Contoured Folding Chairs, 4/Set	15

dtype: int64

Observation

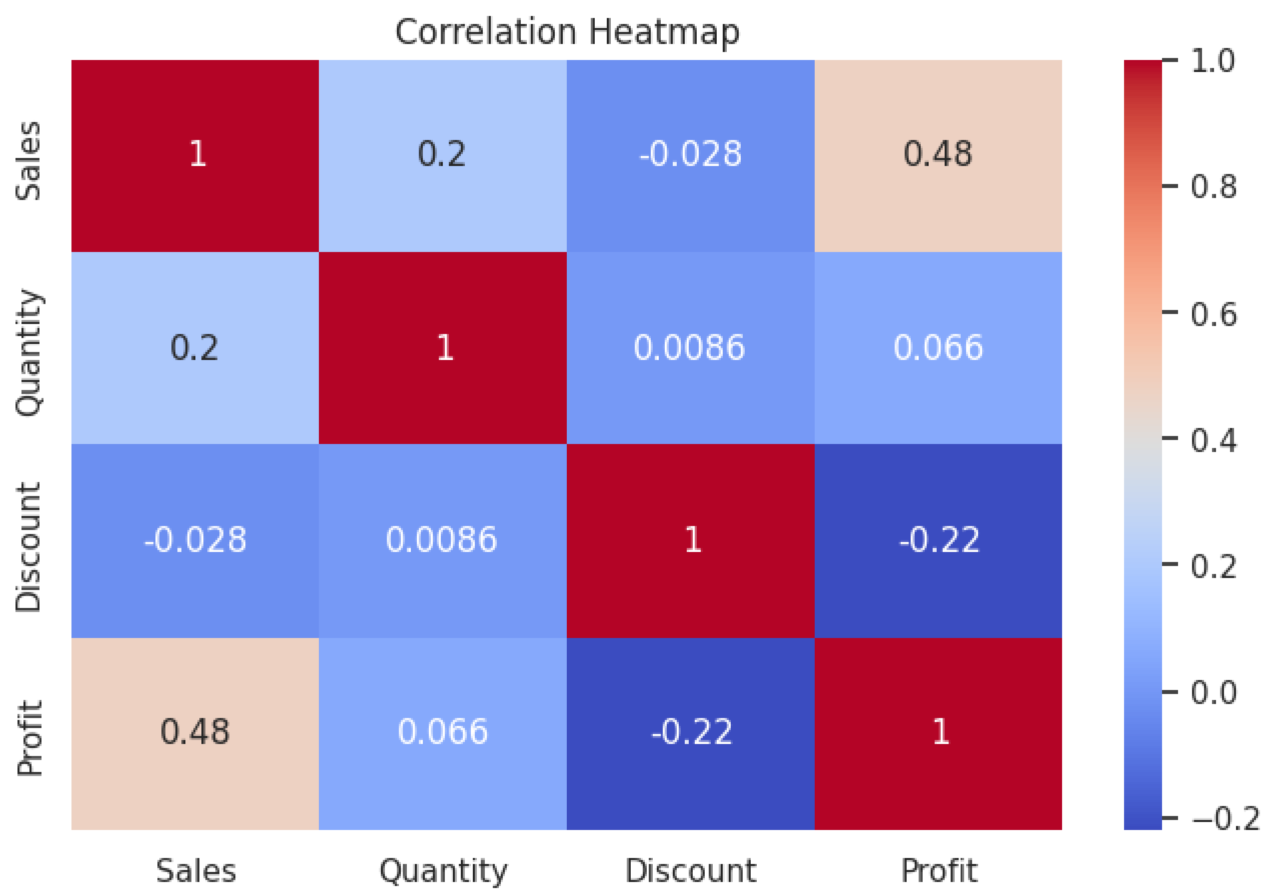
- Standard Class is the most frequently used shipping mode, indicating that customers generally prefer economical shipping options.
- The Consumer segment represents the largest customer group, followed by Corporate and Home Office segments.
- All records belong to the United States, making this a single-country dataset.
- New York City and Los Angeles have the highest number of orders among all cities.
- California, New York, and Texas contribute the largest number of orders among states.
- The West region contains the highest number of transactions, followed by the East region.
- Office Supplies is the most frequently purchased product category.
- Binders, Paper, Furnishings, and Phones are among the most commonly ordered sub-categories.
- Customer purchasing activity is concentrated among a few customers, products, cities, and regions.

```
In [23]:
```

```
plt.figure(figsize=(8,5))

sns.heatmap(
    df[['Sales','Quantity','Discount','Profit']].corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Heatmap')
plt.show()
```



Observation

- Sales and Profit show a moderate positive correlation (0.48), indicating that higher sales generally contribute to higher profits.
- Discount has a moderate negative correlation with Profit (-0.22), suggesting that larger discounts tend to reduce profitability.
- Sales and Quantity have a weak positive correlation (0.20), meaning that larger order quantities contribute somewhat to increased sales.
- Quantity has a very weak relationship with Profit (0.07), indicating that profit depends more on sales value and discount strategy than on quantity alone.
- Overall, Sales and Discount are the most influential factors affecting Profit in this dataset.

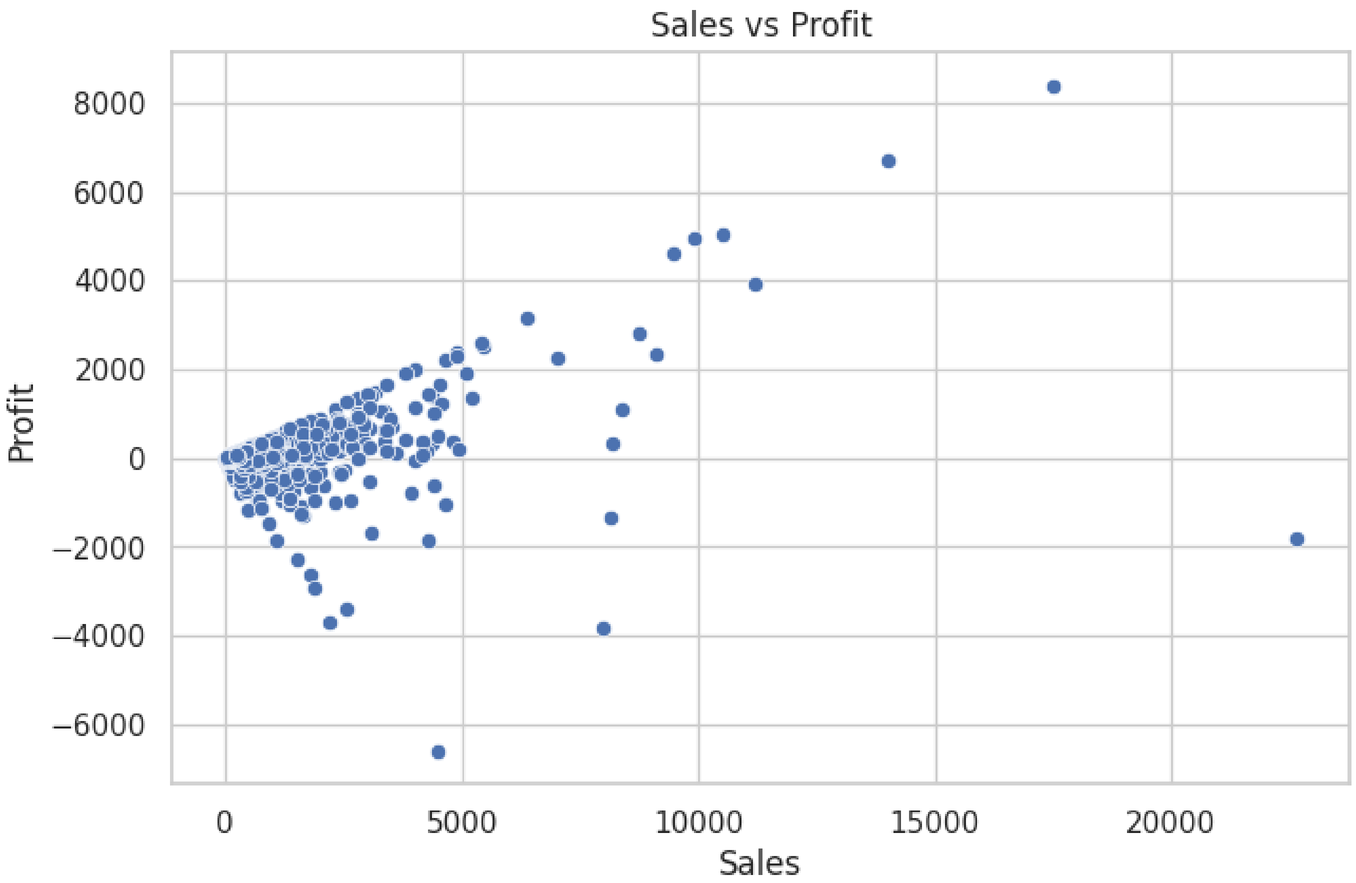
In [14]:

```
for col in df.columns:
    print(col)
```

```
Row ID
Order ID
Order Date
Ship Date
Ship Mode
Customer ID
Customer Name
Segment
Country
City
State
Postal Code
Region
Product ID
Category
Sub-Category
Product Name
Sales
Quantity
Discount
Profit
```

In [15]:

```python
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='Sales', y='Profit')
plt.title('Sales vs Profit')
plt.show()
```

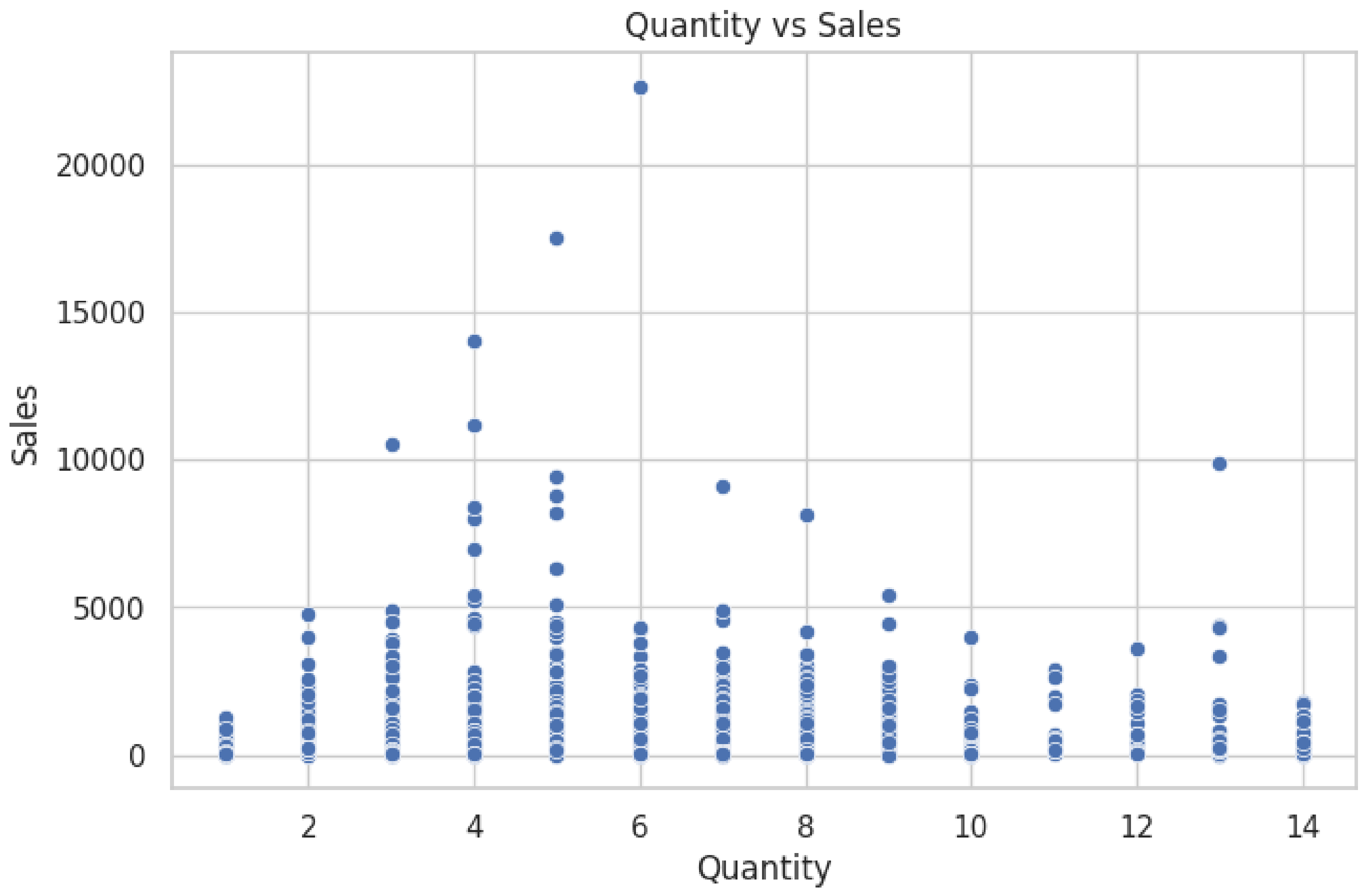


Observation

- Sales and Profit show a generally positive relationship, indicating that higher sales often lead to higher profits.
- Most orders are concentrated in the lower sales range (below 5,000).
- Several high-sales transactions generate substantial profits, reaching over 8,000.
- A few orders with high sales still result in negative profits, suggesting the impact of discounts, costs, or other business factors.
- The presence of both high-profit and high-loss outliers indicates significant variation in order profitability.

In [16]:

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='Quantity', y='Sales')
plt.title('Quantity vs Sales')
plt.show()
```

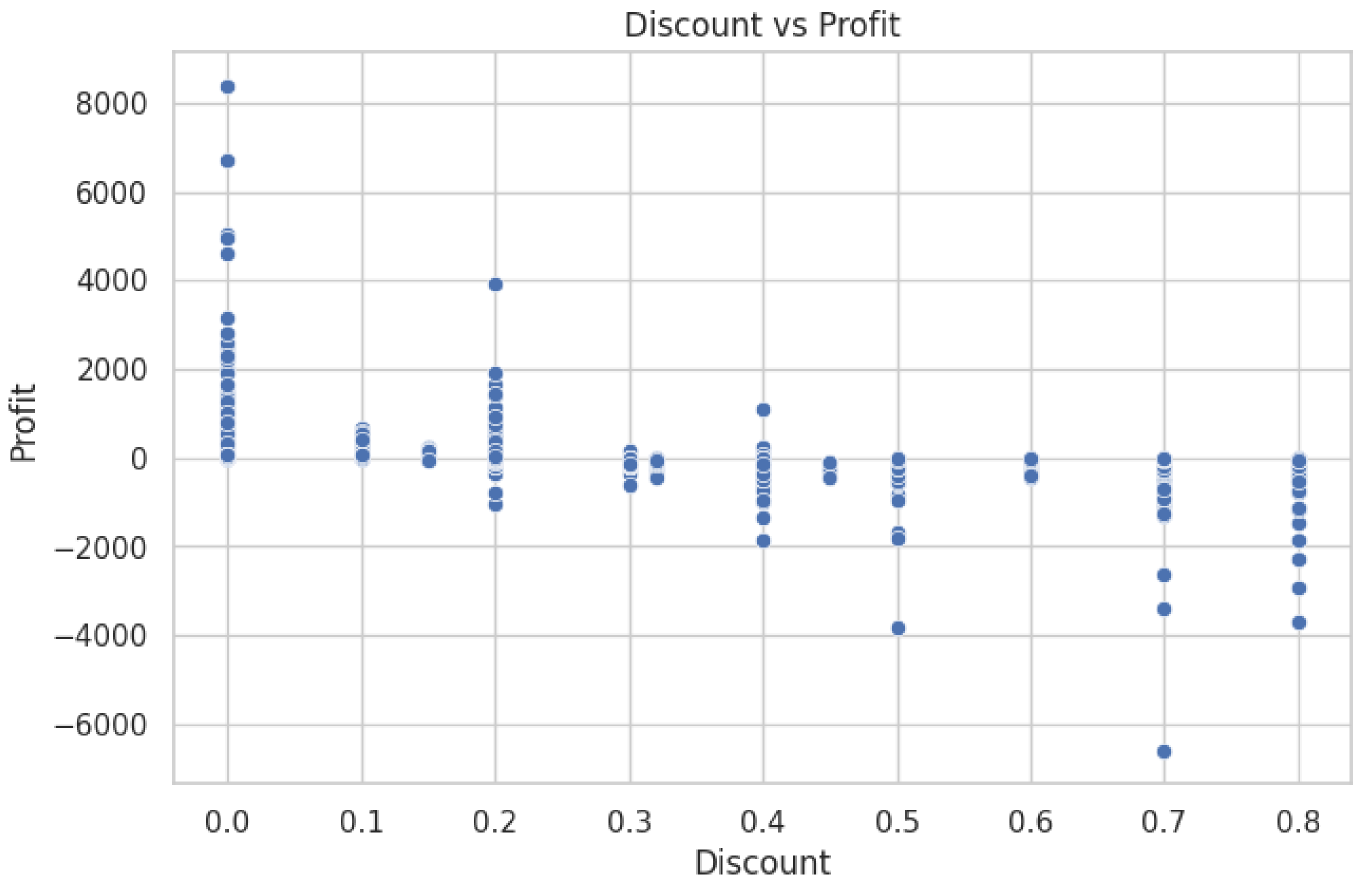


Observation

- Sales generally increase with quantity ordered.
- Most orders involve small quantities with moderate sales.
- A few high-sales outliers are present across different quantity levels.

In [17]:

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='Discount', y='Profit')
plt.title('Discount vs Profit')
plt.show()
```



Observation

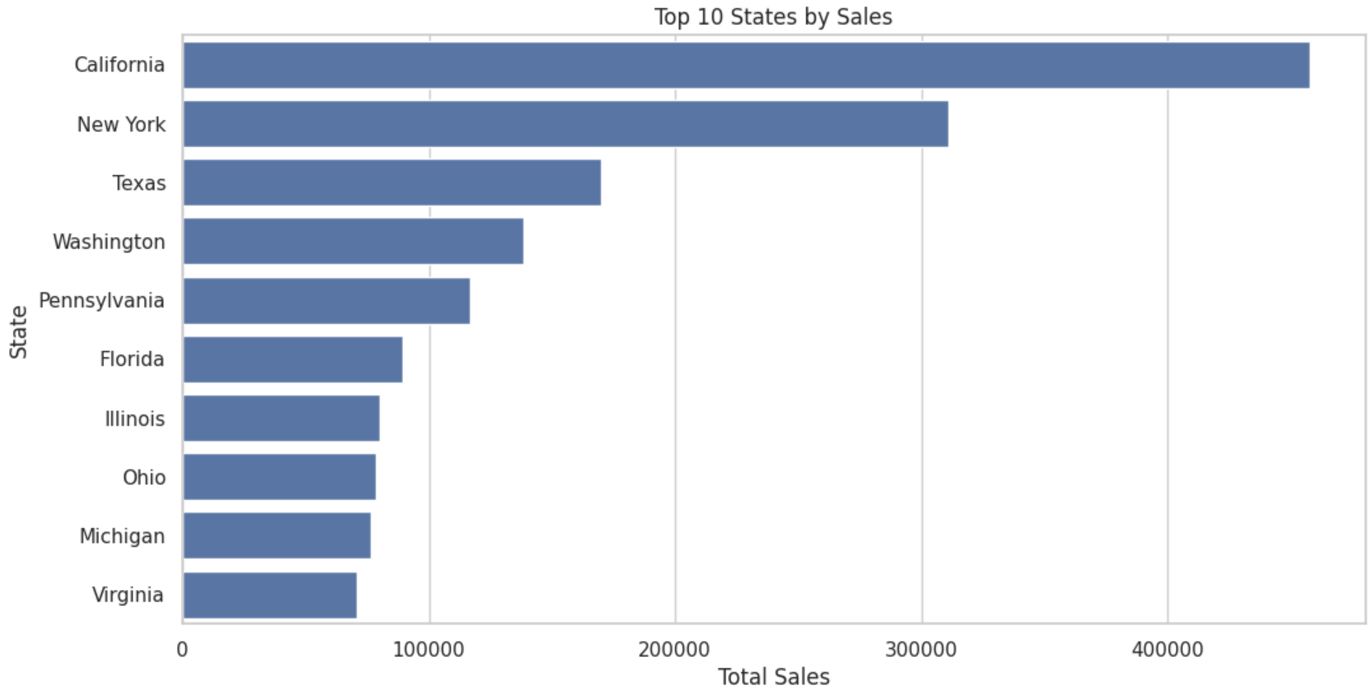
- Profit decreases as discount increases.
- Higher discounts are often associated with lower or negative profits.
- Excessive discounting can significantly impact profitability.

In [18]:

```
top_states = df.groupby('State')['Sales'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(12,6))
sns.barplot(
    x=top_states.values,
    y=top_states.index
)

plt.title('Top 10 States by Sales')
plt.xlabel('Total Sales')
plt.ylabel('State')
plt.show()
```



Observation

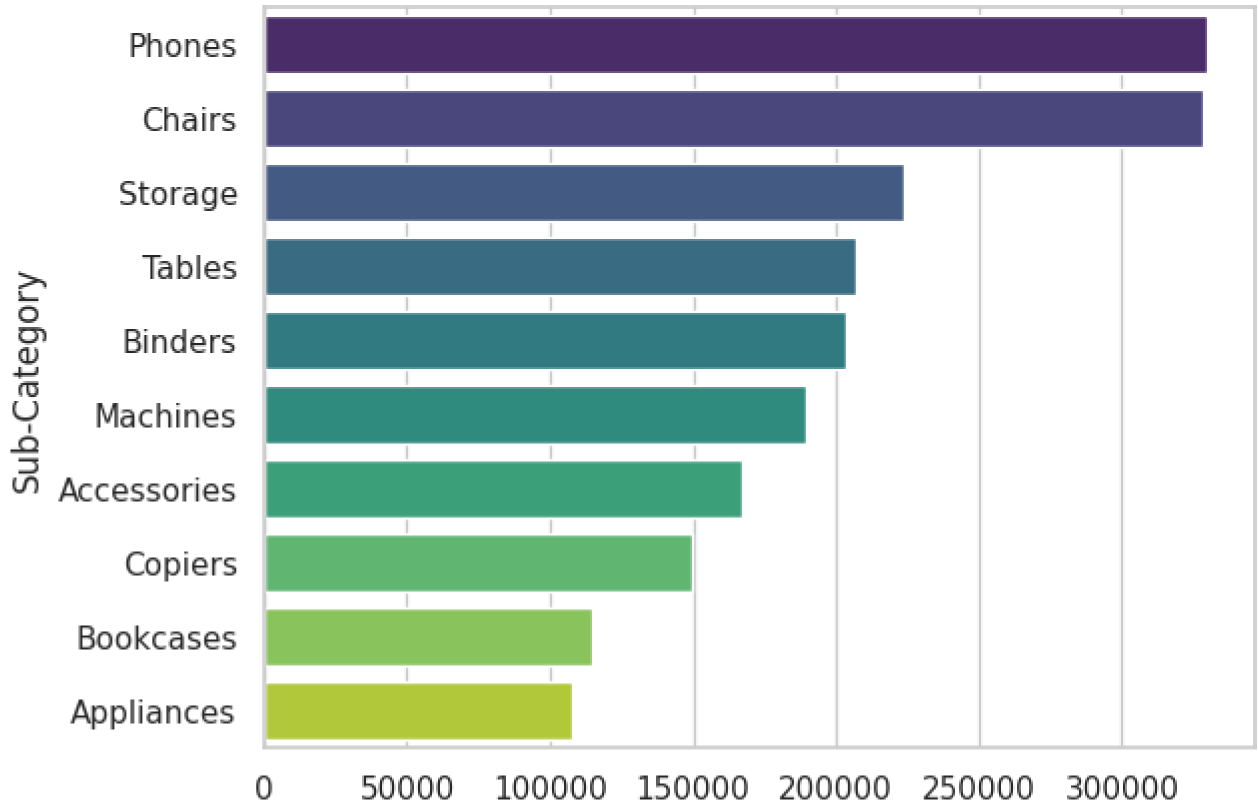
- California generates the highest sales, followed by New York and Texas.
- Sales are concentrated in a few top-performing states.
- California significantly outperforms all other states in total sales.

Observation

California generated the highest sales among all states, followed by New York and Texas. A small number of states contribute a significant portion of total revenue.

In [27]:

```
sns.barplot(
    x=top_sub.values,
    y=top_sub.index,
    hue=top_sub.index,
    palette='viridis',
    legend=False
)
plt.show()
```



Observation

- Phones and Chairs are the highest revenue-generating sub-categories.
- Storage, Tables, and Binders also contribute significantly to total sales.
- Sales are concentrated among a few key product sub-categories.

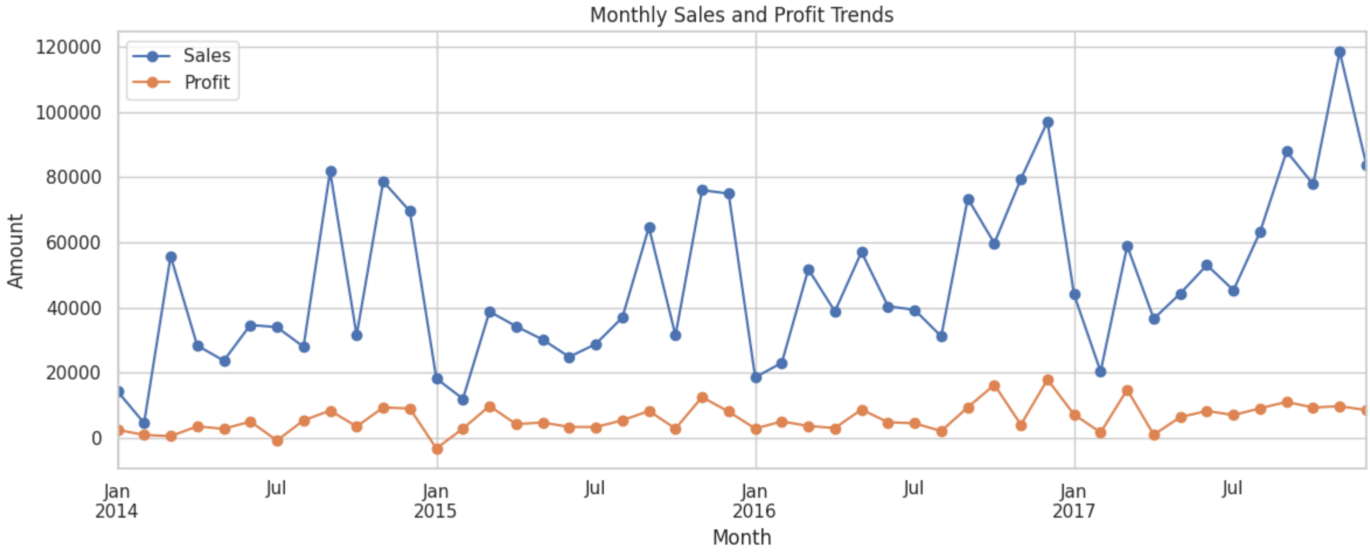
In [24]:

```
# Convert date columns to datetime
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Ship Date'] = pd.to_datetime(df['Ship Date'])

# Calculate monthly sales and profit
monthly = (
    df.set_index('Order Date')
      .resample('ME')[['Sales', 'Profit']]
      .sum()
)

# Plot monthly trends
monthly.plot(figsize=(12, 5), marker='o')

plt.title('Monthly Sales and Profit Trends')
plt.xlabel('Month')
plt.ylabel('Amount')
plt.legend(['Sales', 'Profit'])
plt.tight_layout()
plt.show()
```



Observation

- Monthly sales fluctuate throughout the analysed period.
- Sales generally increase toward the final months of each year.
- Profit follows the sales trend in several months, but high sales do not always produce proportionally high profits.
- Some months show weaker profitability, which may be related to discounts, product mix, or operating costs.

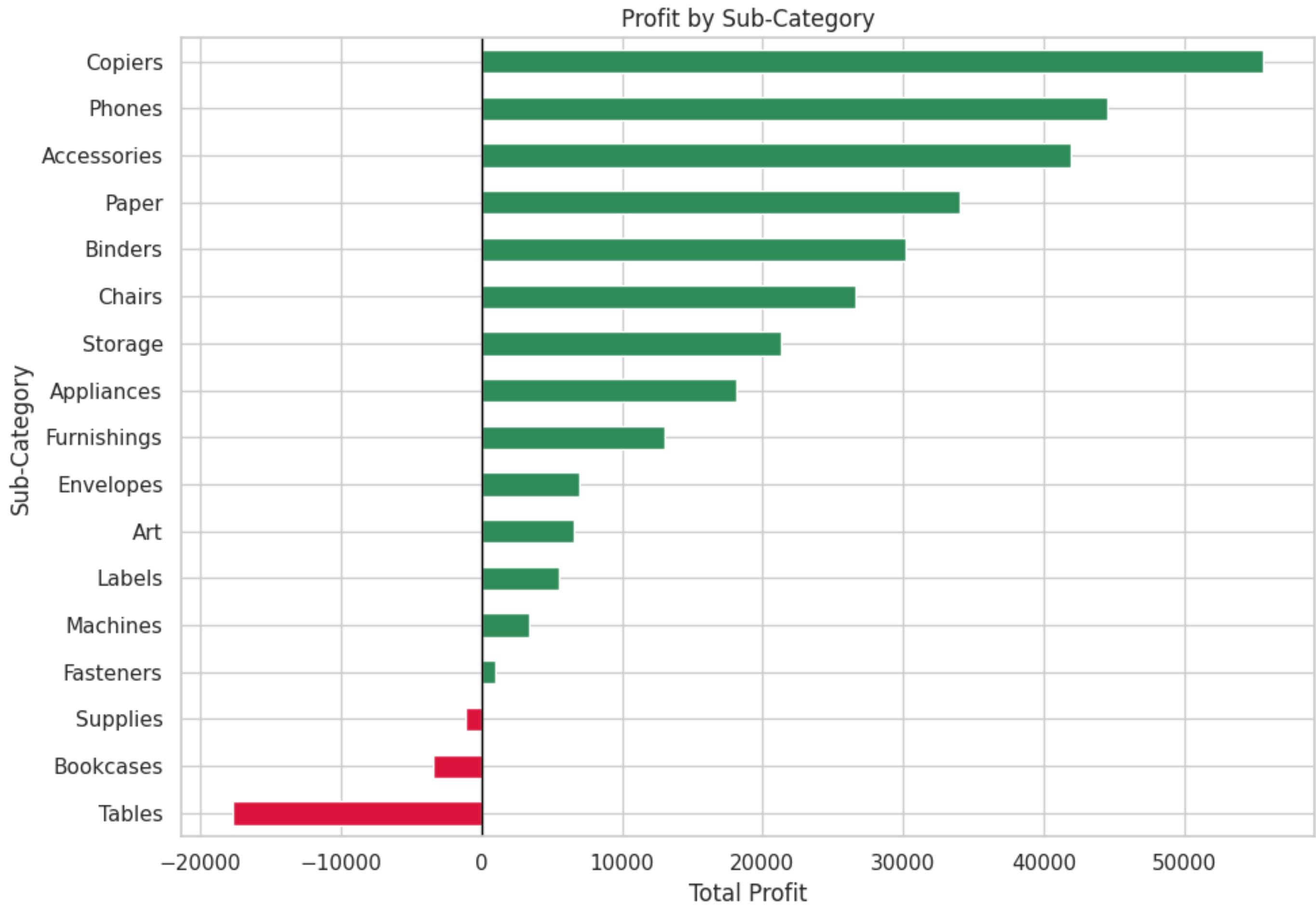
In [25]:

```
subcategory_profit = (
    df.groupby('Sub-Category')['Profit']
      .sum()
      .sort_values()
)

colors = [
    'crimson' if profit < 0 else 'seagreen'
    for profit in subcategory_profit
]

plt.figure(figsize=(10, 7))
subcategory_profit.plot(kind='barh', color=colors)

plt.title('Profit by Sub-Category')
plt.xlabel('Total Profit')
plt.ylabel('Sub-Category')
plt.axvline(0, color='black', linewidth=1)
plt.tight_layout()
plt.show()
```



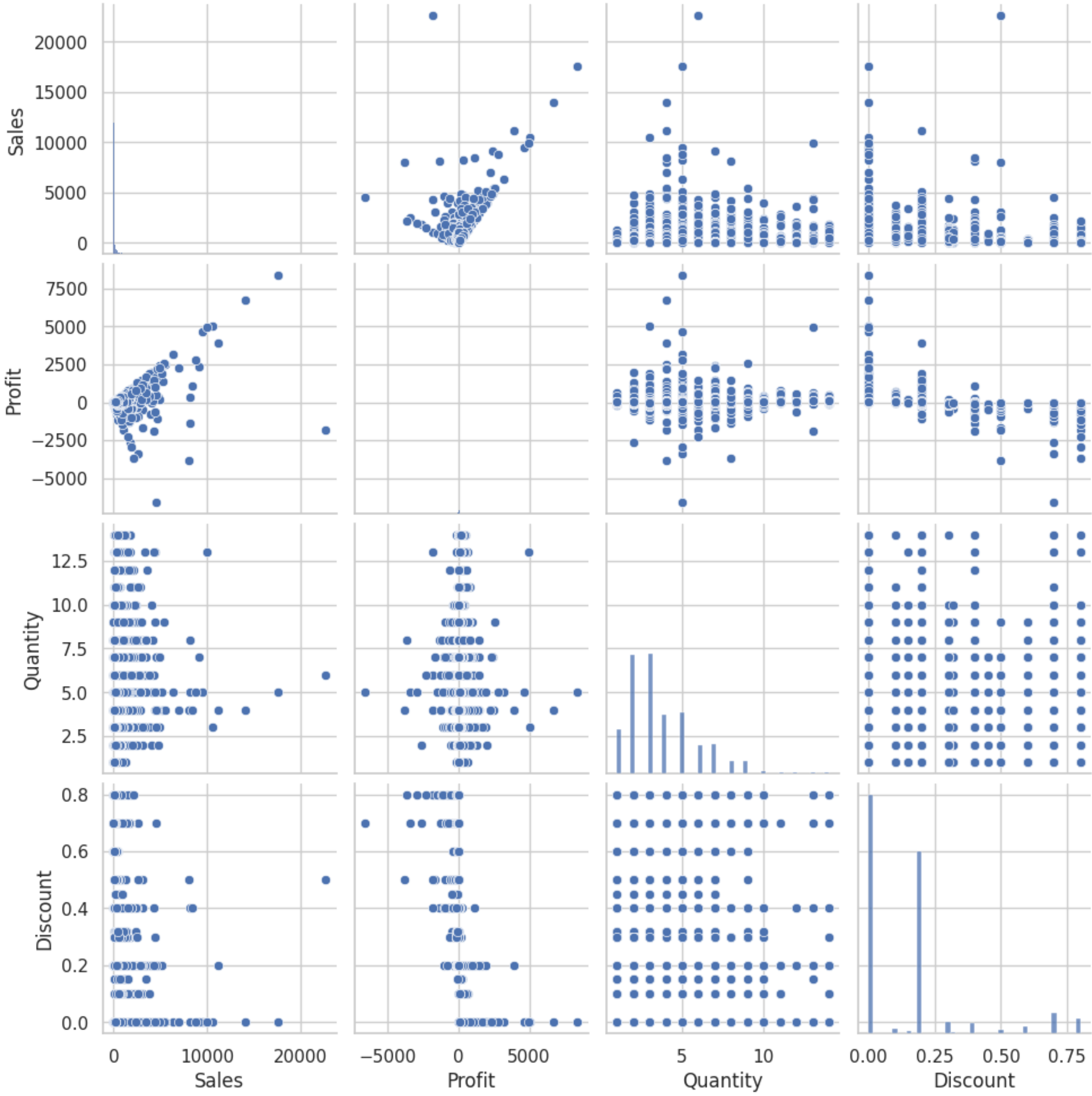
Observation

- Copiers generate the highest total profit, followed by Phones and Accessories.
- Tables produce the largest overall loss despite generating high sales.
- Bookcases and Supplies also show negative profitability.
- High sales do not always result in high profit.
- Loss-making sub-categories may require changes to pricing, discounts, or operating costs.

In [20]:

```
selected_columns = ['Sales', 'Profit', 'Quantity', 'Discount']

sns.pairplot(df[selected_columns])
plt.show()
```

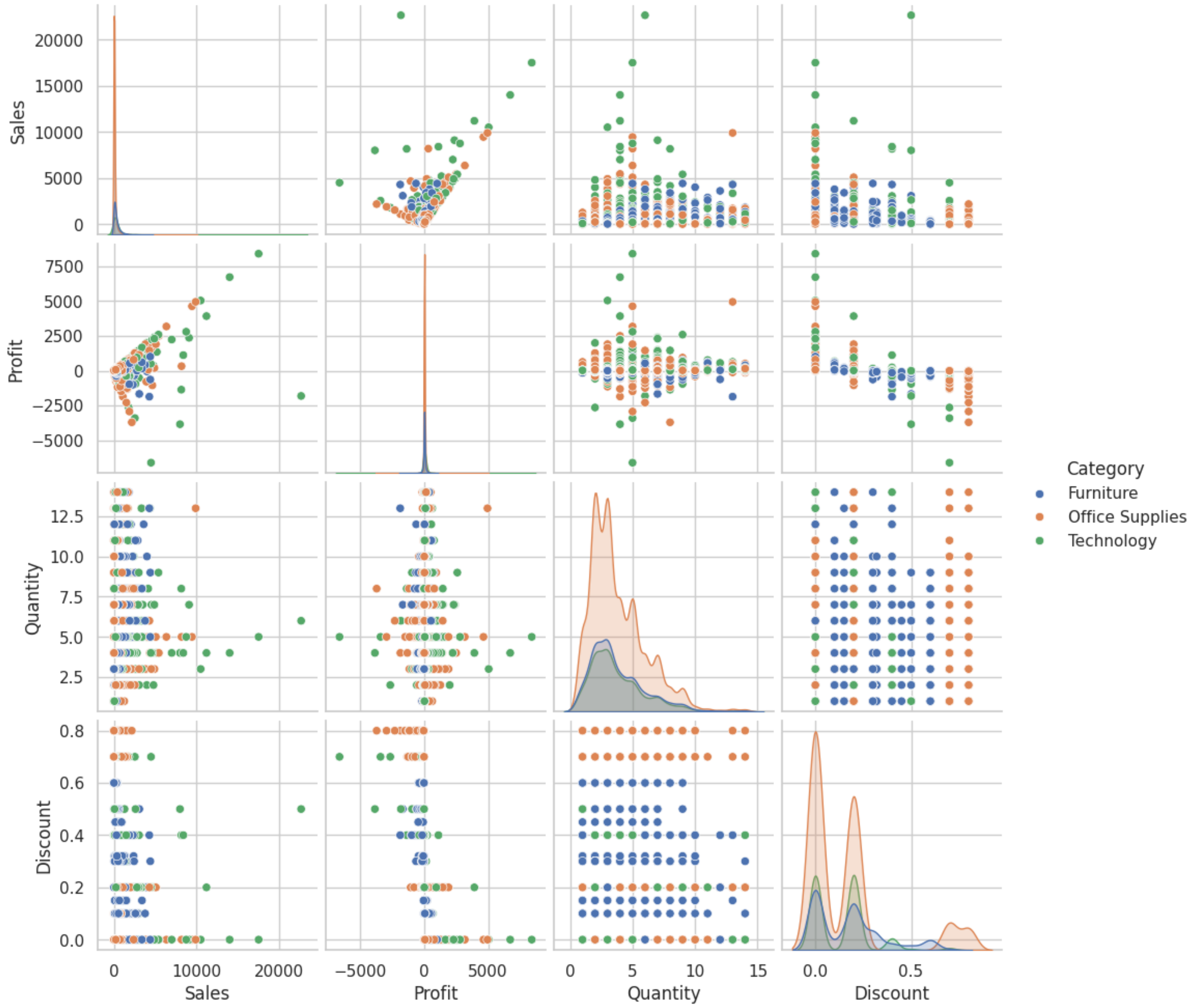


Observation

- Sales and Profit show a positive relationship.
- Higher discounts are generally associated with lower profits.
- Quantity has a weak relationship with both Sales and Profit.
- A few outliers are visible in Sales and Profit.

In [21]:

```
sns.pairplot(
    df[['Sales','Profit','Quantity','Discount','Category']],
    hue='Category'
)
plt.show()
```



Observation

- Technology generates the highest sales and profits.
- Office Supplies have the largest number of transactions.
- Higher discounts tend to reduce profitability.
- Several outliers are present across categories.

In [ ]:

```

```

Final Findings

1. The dataset contains 9,994 rows and 21 columns.

2. No missing values were found in the dataset.

3. The strongest relationship is between Sales and Profit (correlation = 0.48).

4. The most important trend identified is that higher discounts tend to reduce profitability, while Technology products generate the highest sales and profits.

5. Possible anomalies or outliers occur in Sales, Profit, Quantity, and Discount.

6. These results may help with business decision-making, sales optimization, and profit improvement strategies.

Key Insights

1. Technology category generates the highest sales and profit.
2. California is the top-performing state in terms of sales.
3. Phones and Chairs are the highest revenue-generating sub-categories.
4. Higher discounts negatively impact profitability.
5. Sales and Profit show a moderate positive relationship.

Recommendations

1. Focus on high-performing categories such as Technology.
2. Monitor discount strategies to prevent profit loss.
3. Increase marketing efforts in top-performing states.
4. Prioritize high-revenue sub-categories such as Phones and Chairs.

Conclusion

This Exploratory Data Analysis (EDA) identified key sales trends, profitability patterns, customer preferences, and regional performance differences within the Superstore dataset. The findings show that Technology products, California, and top-performing sub-categories contribute significantly to revenue. Additionally, higher discounts were found to negatively impact profitability. These insights can support data-driven business decisions and future predictive analytics initiatives.

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